EnergyWave

energywave.ai • Safely powering AI-driven humanoid robots — a controlled-release posture
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CONFIDENTIAL — FORECAST — FOR DISCUSSION · SUPERSEDES THE AUGUST 2025 PAPER

# **Safe release by design.** If a high-density power source for humanoid robots ever exists, this is how it ships.

In August 2025 a paper on McLean Companies letterhead proposed a safe, government-guided commercial release of a power source for AI-driven humanoid robots. Its purpose stands. Its performance figures, its named business partner and its entity do not: this version carries the purpose forward under EnergyWave's name and claims discipline, and the original stays in the archive, unedited, as a legacy record.

> **What this version withdraws, explicitly.** The 2025 paper stated as fact a 25 kW continuous output, a 3.5 kg / 0.008 m³ package, an energy density "equivalent to 50 MJ/kg", prototypes with "99.98% uptime over 3,000 hours", and expert confirmation that the technology was ready for market. **None of those statements is supported by any document the company holds.** EnergyWave has no prototype, no measured output and no validated energy-conversion mechanism. Every figure in this document is labelled by origin; the only performance numbers are published third-party platform figures, attributed. Any recipient of the 2025 paper receives this version as its supersession.

## 1 · EXECUTIVE SUMMARY

Humanoid robots are a power problem before they are an intelligence problem. To sense and move like a person a platform must run cameras and microphones, chemical sensing, tactile skin and dense actuation at once, and the industry's current answer to endurance is logistics — swap packs, dock, run in shifts — rather than energy. EnergyWave is a research venture built to find out, to a published standard, whether any non-chemical approach can change that; and, separately, to design the control and safety architecture that any high-density portable source would need before it could be released at all. This paper is about the second track. It proposes that the release architecture be developed with government visibility from the start, under binding scope exclusions, so that if the research track ever produces something, the question of safe release has already been answered.

## 2 · THE CONSTRAINT, WITH THE FIGURES ATTRIBUTED

| Statement | Figure | Origin |
|---|---|---|
| Production humanoid battery pack | ≈ 2.3 kWh lithium-ion at ≈ 52 V; next generation keeps the pack and targets efficiency instead | Manufacturer statement via secondary coverage, as cited in v2.0 [1][2]; the dated primary statements are tabulated in v2.1 §1.1 |
| Field runtime under working load | Two to five hours per charge; ninety minutes to two hours in high-activity use on some platforms | Attributed — sector reporting cited in the v2.0 paper [3][4] |

| Statement | Figure | Origin |
|---|---|---|
| Where the load is | Actuation — a potentially major, task-dependent load; holding a position under gravity draws power unless the actuator or transmission cancels or locks the load. Secondary coverage cited by v2.0 puts actuators at ≈ 56 % of one platform's bill-of-materials cost (28 body plus 50 hand actuators) — a cost share, not a power share primary source to be confirmed | Secondary coverage — [2]; qualified as in v2.1 §1.2 |
| Chemistry ceiling | Production cells ≈ 220–271 Wh/kg and packs ≈ 120–200 Wh/kg once housing, cooling and electronics are added, against a cited 500 Wh/kg development goal — a program target, not a physical ceiling; industry planning ≈ 4 % per year, a doubling in about eighteen years | Attributed — as tabulated in DoW White Paper v2.1 §1.3 [13][14][6] |
| Industry response | Autonomous charging stations; inductive charging coils in the feet delivering ≈ 2 kW; high-nickel chemistries accepted with their cost and thermal penalties | Attributed — [7] |
| Energy EnergyWave can deliver today | **None.** | Company statement |

## 3 · WHAT ENERGYWAVE DOES INSTEAD OF CLAIMING

Every candidate mechanism the company examines passes three gates in order — **characterize**, **conservation-test**, **extract or retire publicly** — with pre-registered protocols, replication contracted before each gate, an external advisory board with veto over any performance statement, and negative results published under the company's own name. The first statement of a number, ever, would follow Gate-3 third-party validation. The research program is described in *Research Program and Gates* and is not repeated here. Nothing in this paper depends on that program succeeding.

## 4 · AN ILLUSTRATIVE SCENARIO — ILLUSTRATION, NOT A PRODUCT

> *A household humanoid learns a dish from a kitchen on the other side of the world over a satellite link, checks the pantry, orders what is missing within a budget, and has a hot meal ready when the owner's phone says they are ten minutes from home. Nothing in that scenario is exotic except the power budget: the platform runs human-grade sensing and dense actuation for hours without a dock.*

Kept from the 2025 paper as the kind of continuous, simultaneous load the company means by "a humanoid is a power problem". It asserts no capability. The quantitative version is the Humanoid Power Budget Specification in 03_Technical_Documents.

## 5 · CONTROL AND SAFETY ARCHITECTURE — DEVELOPED ON ITS OWN TRACK

| Principle | Design |
|---|---|
| Platform binding | The module is scoped to ambulatory robotic platforms and cryptographically bound to one host; it authenticates to the platform controller and delivers no output to an unrecognized host — a keyed consumable, like a cartridge and its printer. |
| Scoping in hardware | Power ceiling, ramp rate and duty envelope fixed in hardware to the actuation and compute envelope of an ambulatory platform. Repurposing as a bulk supply means defeating hardware limits — a detectable and destructive act. |

| Principle | Design |
|---|---|
| Tamper-responsive inertion | A tamper-response boundary of the kind standard in high-assurance cryptographic hardware. Envelope breach, unauthorized disassembly, out-of-spec thermal or electrical conditions, or authentication failure trigger an irreversible transition to an inert state — the “paperweight” test. No service mode, no recovery path. |
| Custody and revocation | Units individually identified, attested and remotely revocable. A recovered or diverted unit is a dead unit. |
| Compartmentalization | Know-how and any future hardware held in imageable / process / state bins; split manufacturing; post-quantum platform binding — Compartmentalization & Anti-Tamper Posture v0.1 (Sept 2026), counsel questions in its §8 still open. |

**Negative scope — offered as binding conditions of any government relationship, not as a statement in a paper.** No directed-energy application. No propulsion application. No munitions or ordnance integration. No general-purpose or grid-scale power configuration. No export or foreign transfer absent explicit U.S. Government authorization. The 2025 paper’s own exclusions — “excluding laser cannons or exotic propulsion” — are kept and widened.

## 6 · RELEASE ROADMAP — EVERY STAGE CONTINGENT ON THE GATE BEFORE IT

| Stage | Condition to enter | Work |
|---|---|---|
| **Stage 0 — now** | None; standalone value for any high-density portable source, including conventional ones | Control architecture specified and adversarially tested independent of any energy result; disclosure framework discussed with the relevant program offices; scope exclusions written into an agreement. |
| **Stage 1 — safety validation** | Gate 3 passed: independently validated, repeatable performance under third-party observation | Government-witnessed verification of stability and safeguards on the actual module; platform integration tests; encrypted control systems in place. The 2025 paper’s “0–12 months” is replaced by this condition; no date is stated. |
| **Stage 2 — controlled pilot** | Stage 1 report accepted | Deployment with vetted commercial partners under the safety standard agreed in Stage 1; user guidelines; infrastructure compatibility. |
| **Stage 3 — scaled release** | Stage 2 outcomes published | Production under regulatory oversight; distribution limited to authorized partners; the negative scope above remains binding. |

## 7 · WHY GOVERNMENT COLLABORATION, AND WHY NOW RATHER THAN LATER

**Safety validation.** An energy-density improvement of the magnitude that would matter is dual-use by construction. The appropriate time to establish government visibility is before results exist, not after.

**Proliferation control.** Custody, revocation and scope restriction are easier to design in than to retrofit; the company would rather be inside a framework early than negotiate one under pressure.

**A disciplined negative is also a deliverable.** If the research track retires every candidate, the control architecture, the measurement platform and the published negatives remain, and the question is closed in public rather than recirculated for another decade.

## 8 · ENTITY AND IP STATUS — STATED AS IT STANDS

| Item | Status |
|---|---|
| Issuing entity | EnergyWave, Inc. (Delaware C corporation) — formation in progress with K&L Gates LLP; the file number and date are added on formation. The 2025 paper's issuer, McLean Companies LLC, is the founder's existing company and is not the issuer of this document. |
| Provisional application 63/763,998 | Filed 27 February 2025 in the founder's name; expired after twelve months without a non-provisional filing. Counsel's path: a coversheet provisional in the inventors' names, assigned to the company on formation. The 2025 paper's "backed by provisional patent" is withdrawn. |
| Federal record | xTech Search 10 entry submitted 9 October 2026 by McLean Capital Development LLC (Wyoming), the applicant of record; see the Submission Summary. |
| Named persons | The team is as named on the 13 October deck — founder, co-founder, a team member (Jack Sarfatti — theoretical physicist), an advisor (Alan Crunkleton), patent counsel — with Mike Wentz a team member (role to be announced) and Arath Acosta a planned Design Engineer from the Seed Round; details in *Team and Counsel*. Persons named in the 2025 paper who are not on that roster are not named here. |

## 9 · THE ASK

A technical review meeting on four questions: whether the ambulatory-platform power constraint in §2 is recognized as a capability gap within the relevant program offices; whether a conservation-tested characterization program with public retirement criteria is of interest as a low-cost, high-information investment; whether the control architecture in §5 should be developed under Government specification, independent of the energy research outcome; and what framework would govern disclosure, custody and scope restriction should the research ever produce a positive result. The company is not asking for validation of a claim.

**EnergyWave, Inc.** (Delaware — formation in progress) · a McLean Companies venture · P.O. Box 8646, Newport Beach, CA 92658
Ian McLean, Founder & CEO · ian@energywave.ai · +1 (949) 280-0936 · energywave.ai

This document contains no classified information and no export-controlled technical data. It describes no device design, fabrication method or operating parameter. Early-stage research venture; forward-looking statements are not proven capabilities or guarantees.

Basis: the August 2025 paper (superseded); the August 2026 white paper v2.0 §§1, 3–5, 7 and its sources; the Compartmentalization & Anti-Tamper Posture v0.1; the 13 October 2026 deck; the Corporate, IP and Legal Status and Team and Counsel pages. Platform figures are attributed published statements, not company measurements.